

Cassava's harvest residue biochar from rotary kiln pyrolysis: Effects on sandy loam soil properties and maize growth

Author links open overlay panelWenes Ramos Silva ^a, Roberta Menezes Santos ^b, [Alberto Wisniewski Jr ^a](#)

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Highlights

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600 °C produces a biochar with higher ash content, aromaticity, and surface area.

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Biochars bring K > P > Ca elements as a plant nutritional complement.

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Cassava biochar increases the pH of sandy loam soil.

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50-ton ha⁻¹ of biochar increased soil saturation by 10–18 % on SWRC.

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50-ton ha⁻¹ of 600 °C cassava's biochar increased soil water holding capacity by 20 %.

Abstract

Cassava shoot system (CSS) biomass is a smart and despised raw material for biochar production. Commonly, cassava roots and their husk are most exploited as feedstock for the thermoconversion process. In this research, we proposed the production of biochar from CSS in a continuous pyrolysis reactor at temperatures of 400 °C, 500 °C, and 600 °C, searching for added value to this agricultural residue and to produce a soil amendment material with good water retention potential. CSS-biochars have their effects tested on sandy loam soil at 25- and 50-ton ha⁻¹ application rates. Considering soil amendment and carbon sequestration purposes of biochars, the volatile matter, fixed carbon, and ash content are important parameters to be considered. Higher temperature produces more

stable and less organic functionalized biochars, measured by the O/C values from 0.21 to 0.03 and H/C from 0.96 to 0.46. Biochars' pH increases directly with the ash content and significantly alters the soil pH, increasing it from 5.0 to 6.0 with BC500C and BC600C at a rate of 50 tons ha⁻¹. At the same application rate, BC400C and BC600C presented an increase in water-holding capacity of 13 % and 20 % compared with the control sample, respectively. Overall, the matric potential view for all application rates only the CSS-biochar produced at 600 °C demonstrated superior water retention compared to the reference control. An important observed aspect is the no significant impact of biochar on the maize germination step in pot experiments.

Graphical abstract



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Introduction

Due to its nutritional and economic importance, cassava is a plant grown in more than 100 tropical and subtropical countries worldwide [1]. Among South American countries, Brazil was the largest producer of cassava in 2022, with a total of approximately 17.6 million tons produced, followed by Paraguay, with a total production of approximately 2.6 million tons [2]. Since the root is the desired part of the cassava plant, harvesting involves uprooting the entire plant and separating the roots. After harvesting, the remaining parts of the plant—such as stems, rhizomes, and leaves—are usually left in the field. This remaining vegetation, known as harvest residue, typically represents about half of the total biomass produced during cultivation. On average, harvest residue accounts for approximately 63 % (on a dry basis) of the weight of the harvested roots [3].

In reusing waste biomass, pyrolysis has emerged as a promising route for converting these materials into economic value-added and energetic products [4]. The pyrolysis process consists of the thermal degradation of materials in the absence of oxygen, which results in

obtaining solid, liquid, and gaseous products. The yield and chemical composition of the products depends mainly on the type of biomass used and the pyrolysis conditions employed, such as residence time, temperature, and heating rate [5]. The solid carbonaceous product of pyrolysis, named biochar, is characterized by its porous structure, high carbon content, and high surface area. Its composition comprises aromatic carbon structures with a surface rich in functional groups [6]. Due to its chemical characteristics, biochar has been tested as an adsorbent of organic [7] and inorganic contaminants in water [8] and soils [9]. Furthermore, biochar soil application has been considered an efficient carbon sequestration method [10,11], as well as a product for improving soil quality and fertility [11,12].

The application of biochar as a soil amendment is gaining significant attention and demonstrating promising results. Santos et al., 2022 [13], tested biochars from green coconut husks, sewage sludge, and sugarcane bagasse in different tropical soils. The study demonstrated that these biochars increased the pH, organic matter content, and electrical conductivity of the soils. They also improved soil porosity and water transmission, which increased soil water retention. Zhang et al., 2025 [14], investigated the effect of the application of different biochars in sandy loam soil and demonstrated positive effects on lettuce growth. The results showed an increase in plant mass by 62 %–101 % with the addition of 3 %–5 % biochar to the soil. Gonzaga et al., 2025 [15], studied the use of green coconut biochar in soil and found that pure biochar enhanced soil pH, raised cation exchange capacity, base saturation, and concentration of inorganic nutrients. Furthermore, biochar pretreated with poultry litter promoted better growth in corn plants, increasing both aboveground and root biomass. Wijitkosum and Sriburi, 2021 [12], found that the application of cassava biochar improved soil properties for corn cultivation, yielding better results in conditioned soil. Additionally, they demonstrated that using fertilizers in conjunction with biochar optimizes the effects of the fertilizers, resulting in better outcomes than using fertilizers alone. It demonstrates that biochar can enhance soil quality and reduce water stress, which consequently increases crop fertility and productivity [16], especially in tropical soils, where it improves nutrient availability and corrects pH levels [17].

Based on this, it was noted that biochar production from agricultural waste is well established and its benefits to the soil have been well documented. However, the main production route for these biochars studied is batch pyrolysis on a bench scale. Thus, the study of biochar production in a continuous pyrolysis reactor is investigated here for the first time, in which we use the cassava sprout system as raw material for a commercial-scale process for the production of biochar for agricultural purposes, along with the effects of its application on soil hydrological properties. Therefore, this study demonstrates the

production of biochar through a continuous pyrolysis process in a rotary kiln reactor. The biochars produced were assessed for their chemical composition, structural characteristics, and impact on soil properties. The study focused on how cassava biochar influences the water retention characteristics of sandy loam soil, with an emphasis on identifying the optimal production temperature and application rate. The data from this work demonstrate the biochar production scalability through a continuous pyrolysis process to obtain biochar for agricultural purposes and fill gaps not previously addressed in the literature for this residual biomass.

Biochar preparation and chemical characterization

The biochars were produced at an adapted semi-pilot-scale rotary kiln reactor (model FRO-1100, ForteLab Ltda.) featuring a cylindrical stainless-steel furnace with diameter of 75 mm and length of 500 mm, with inclination and rotation control, as described by Santana et al., 2020 [18]. Our previous work describes the collection, processing, chemical characterization, and continuous pyrolysis process conditions of the cassava shoot system (CSS) biomass [19]. In this process, the biomass was feed

Immediate and elemental analysis

It was observed in Table 1 that as the temperature of the pyrolysis process increased the moisture and volatile material contents decreased, while the percentages of fixed carbon and ash in the biochars increased. This behavior was also noted by Zhai et al., 2024 [23], in their analysis of tropical crop residues biochars produced at different temperatures (200, 300, 400 and 500 °C) using slow pyrolysis in a fixed bed reactor. The elemental composition of biochars also varied with pyrolysis

Conclusions

This study evaluated the composition and soil application of CSS biochars produced at different temperatures using a continuous pyrolysis process in a rotary kiln reactor. Analysis of the solid pyrolysis product revealed that the thermoconversion temperature influenced the chemical characteristics of the biochar. Increasing the pyrolysis temperature resulted in higher degradation of the feedstock, leading to a lower degree of oxygenated functionalization in the biochars and increased

CRedit authorship contribution statement

Wenes Ramos Silva: Writing – original draft, Investigation, Formal analysis, Data curation. **Roberta Menezes Santos:** Investigation, Formal analysis, Data curation. **Alberto Wisniewski Jr:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

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Competing interests

The authors have no competing interests to declare that are relevant to the content of this article.

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